Quinn O. Casey

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RESEARCH INTERESTS

Active galactic nuclei (AGN) • galaxy formation & evolution • AGN variability and feedback • spectroscopy & IFUs • multi-messenger astronomy • starburst-AGN connection • low-mass galaxies • resolved stars • big data & ML

EDUCATION

Ph.D., Physics and Astronomy — Expected 2028
Advisor: Dr. Ryan Hickox, Dartmouth College

B.S., Physics and Astronomy — May 2021
Advisor: Dr. Christy Tremonti, The University of Wisconsin-Madison

TEACHING EXPERIENCE

Teaching Assistant, Department of Physics and Astronomy, Dartmouth College

Foreign Study Program: Cape Town, South Africa	Winter 2027
ASTR 61: Observational Techniques in Astronomy (w/Lab)	
ASTR 81: Special Topics in Astronomy	
PHYS 31: Research Methods in 21st Century Astronomy in South Africa	
ASTR 02: Exploring the Universe	Summer 2026
ASTR 19: Habitable Planets	Sping 2026
PHYS 04: General Physics II (w/Lab)	Spring 2025
ASTR 02: Exploring the Universe (w/Lab)	Fall 2024
ASTR 19: Habitable Planets	Spring 2024
ASTR 01: Exploration of the Solar System (w/Lab)	Summer 2023
ASTR 04: Development of Astronomical Thought	Spring 2023
PHYS 13: Introductory Physics I (w/Lab)	Winter 2023

Teaching Assistant, Department of Astronomy, UW-Madison

ASTR 103: The Evolving Universe: Stars, Galaxies, and Cosmology	Fall 2021

OBSERVING EXPERIENCE

MDM Observatory — 2023 - 2025
Hiltner Telescope, *CHαS* (15 nights)
Hiltner Telescope, *OSMOS* (5 nights)
McGraw-Hill Telescope, *Templeton Direct* (10 nights)

Las Campanas Observatory — Dec 2023
Magellan Clay Telescope, *MEGACAM* (2 nights)

OBSERVING AND ARCHIVAL PROGRAMS

Awarded as PI .....

MDM Observatory/*CHαS*
Extended AGN Emission in Nearby Galaxies 15 nights, 2024

Awarded as Co-I .....

Roman Space Telescope Cycle 1 (PI: R. Hickox)
Red Dots at Cosmic Noon with Roman Archival, 2026

South African Large Telescope/*NIRWALS* (PI: R. Hickox)
NIR Spectroscopy of Luminous Obscured AGN 10.4 hours, 2024-2
NIR Spectroscopy of Dwarf AGN Candidates 10 hours, 2024-2

FUNDING, HONORS, & AWARDS

The Selamawit Tsehaye Teaching Award Spring, 2025
Dartmouth College

Matt Johns Funding Award ($3600) Spring, 2024
Dartmouth College

NSF GRFP Honorable Mention 2024

Lowell Doherty Award for Excellence in Astronomy Spring, 2021
UW-Madison

CONFERENCES, PRESENTATIONS, AND WORKSHOPS

- Lisa Kewley's group meeting, Harvard University November, 2026
- Supermassive Black Holes and Blue Notes, Université de Montréal July, 2026
  Contributed Talk: *A Population of Local Red Dots in SDSS*
- CosmicAI: AstroAI Boot Camp, Austin TX June, 2026
- Galactic Fronteirs II (**LOC**), Dartmouth College July, 2025
  Poster: *Deep Photometric Observations of Ultrafaint Milky Way Satellites Centaurus I and Eridanus IV*
- 244th Meeting of the American Astronomical Society, Madison, WI June, 2024
  Poster: *Using Extended Emission to Uncover Past AGN Activity with MaNGA*
- What Drives the Growth of Black Holes (**LOC**), Reykjavik, Iceland September 2022
- 239th Meeting of the American Astronomical Society, Virtual January, 2021
  Poster: *Low-Metallicity AGN in eBOSS*

COLLABORATIONS

Roman Science Collaboration Fall 2025 - Present
LSST AGN Working Group Fall 2024 - Present
DECam Local Volume Exploration Survey (DELVE) Fall 2022 - Fall 2025

SKILLS

Data & modeling: Markov Chain Monte Carlo (`emcee`), forward modeling, high-performance computer clusters
Techniques: SED-fitting, `GALFIT`, `DAOPHOT`
Fluent: python, LaTeX, git, IDL, ADQL (SQL92)
Packages: pandas, NumPy, Matplotlib, Astropy, SciPy, sklearn, PyTorch, TensorFlow
Experience with: html, Bash, R, SQL, iraf, pyraf, virtual machines

MENTORSHIP, OUTREACH, & SERVICE

Graduate Academia Survival Program (GASP) — Summer 2024 – Present
Program founder, coordinator, and peer-mentor
Dartmouth College, Department of Physics and Astronomy

Graduate Assembly Chair — Summer 2024 – Present
Advocate on behalf of the physics & astronomy graduate student body
Manage budget, liaison w/Chair, grad recruitment, conflict mediation, curriculum planning
Dartmouth College, Department of Physics and Astronomy

PENGUIN — Fall 2026 – Present
Peer-mentor
Various Institutions

Graduate Liaison to the DEI Committee — Fall 2023 – Spring 2025
Dartmouth College, Department of Physics and Astronomy

Montshire Astronomy Day — 2023 - 2026
Montshire Museum of Science

Solar Eclipse Viewing — Spring 2024
St. Johnsbury, VT

PUBLICATIONS

Summary (2026-11-06): ADS search • 7 refereed • 3 lead author • 57 citations • h-index 5 .....

- *AGN Archaeology: A Systematic Search for Fading AGN in MaNGA*
  Q. O. Casey, R. Hickox, J. Cohn, and 8 additional authors
  The Astrophysical Journal (in press)
- *A Population of Little Red Dot-like Quasars in SDSS*
  Q. O. Casey, R. Hickox, N. Cleri, and 6 additional authors
  The Astrophysical Journal (2026), 1009, 220C
- *Deep Photometric Observations of Ultrafaint Milky Way Satellites Centaurus I and Eridanus IV*
  Q. O. Casey, B. Mutlu-Pakdil, D. Sand, and 20 additional authors
  The Astrophysical Journal (2025), 984, 148C
- *Evidence for Evolutionary Pathway-dependent Black Hole Scaling Relations*
  J. H. Cohn, E. Durodola, **Q. O. Casey**, and 2 additional authors
  The Astrophysical Journal Letters (2025), 988L, 61C
- *Off-Nuclear AGN Signatures in MaNGA: Spatially resolved selection using MEGACUBES*
  T. Brosi, D. Wylezalek, M. Albán, and 5 additional authors (including **Q. O. Casey**)

Submitted to Astronomy & Astrophysics (2026)

- *Kinematic Signatures for Fossil AGN Outflows in the SDSS MaNGA Sample*
  S. Jaison, R. Riffel, D. Wylezalek, and 11 additional authors (including **Q. O. Casey**)
  Submitted to the Monthly Notices of the Royal Astronomical Society (2026)
- *Hubble Space Telescope Imaging of Three Isolated Faint Dwarf Galaxies Beyond the Local Group: Pavo, Corvus A, and Kamino*
  B. Mutlu-Pakdil, M. Jones, D. Sand and 14 additional authors (including **Q. O. Casey**)
  The Astrophysical Journal (2026), 997, 271M
- *Deep Imaging of Grus II and Horologium II: Structure and Extent of Two Ultra-Faint Milky Way Satellites*
  D. Prabhu, D. Sand, A.Chiti and 29 additional authors (including **Q. O. Casey**)
  The Astrophysical Journal (2026), 1010, 150P
- *DELVE Milky Way Satellite Census I: Satellite Population and Survey Selection Function*
  C. Tan, A. Drlica-Wagner, A. Pace, and 84 additional authors (including **Q. O. Casey**)
  The Astrophysical Journal (2026), 1000, 87T